

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 8 Summer 2025 (Regular)

Course : Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTMEC801A - FUNDAMENTALS OF AUTOMOTIVE SYSTEMS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

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- 1 Which are the most important requirements of an automobile carburetor?
- a. ensuring full torque at low speeds
 - b. good fuel economy
 - c. absence of racing of the engine under idling conditions
 - d. All of these

What is the combustion in spark ignition engine?

- a. heterogeneous
- b. laminar
- c. homogeneous
- d. none of these

- 3 Gasoline engines are _____ than diesel engines.
- a. lighter
 - b. unpredictable
 - c. heavier
 - d. none of these

The timing of injection has to be advanced as the speed _____

- a. increases
- b. stables
- c. decreases
- d. All of these

_____ in supercharging pressure increases the tendency to detonate and pre-ignite.

- a. Decrease
- b. Increase
- c. Unpredictable
- d. none of these

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- 6 Which of the following factors affecting combustion in the CI engine?
- intake temperature
 - jacket water temperature
 - compression ratio
 - All of these
- 7 In a disc brake, which component provides the pad-to-disc adjustment?
- Bleed screw
 - Piston
 - Caliper
 - Piston seal
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- Generally which brakes are on the Rear wheels?
- Drum brake
 - Disk brake
 - Shoe brake
 - Double shoe brake
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- 9 If the front of the front wheels is inside and rear of front wheels are apart when the vehicle is at rest, then the configuration is called?
- Toe-in
 - Toe out
 - Positive camber
 - Positive castor
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- What is the name of the angle through which the wheel has to turn to sustain the side force? <https://www.batuonline.com>
- Slip angle
 - Castor angle
 - Camber
 - Kingpin inclination
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- Which of these were or are used in automobiles to provide suspension.
- Coil springs
 - Torsion bars
 - Leaf springs
 - All of these
- 12 Which type of wheels is preferred in sports cars?
- Disc wheel
 - Wire wheel
 - Magnesium alloy wheel
 - Aluminum alloy wheel
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- Q2. Solve the following.
- A) What is a cam shaft? Discuss the various functions of a camshaft in an IC engine. 6
- B) What is engine knocking? What are the factors that influence knocking in SI engines? 6
- Q3. Solve the following.
- A) Explain the working of a synchromesh gear box used in Automobiles, with neat sketch. 6
- B) Explain Construction and working of Multi Plate clutch with sketch. 6
- Q4. Solve Any Two of the following.

- A) What is supercharging? How it is done explain with necessary sketch. 6
- B) What is an engine block? Discuss the functions of wet sleeve and dry sleeve inserts with neat sketches 6
- C) Explain emission control systems used in IC engine. 6
- Q5. Solve Any Two of the following.
- A) Explain with neat sketch the components and working of a Drum Brake. 6
- B) With neat sketch explain working of internal expanding brake and enlist its advantages. 6
- C) Why a steering is required in an automobile? Explain Re-circulating ball type steering gear with necessary sketch. 6
- Q6. Solve Any Two of the following.
- A) Explain with neat sketch working of Macpherson Strut Suspension system. 6
- B) Differentiate between Tyres with tube and Tubeless tyres. 6
- C) What is ABS? Discuss with a neat sketch the operation Methodology of ABS. 6

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